Design of a Two-Stage Operational Amplifier

Progetto di circuiti e sistemi analogici M

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1. Tables

Table 1: Project Specs

	V_{DD}/V_{SS}	SR	C_L	$V_{in,CM}^{max}$	$V_{in,CM}^{min}$	V_o^{max}	V_o^{min}	f_{0dB}	DC_{gain}	PM	$\frac{v_{in,N}(f)}{\sqrt{\Delta f}}$
Value	$\pm 2.5V$	$\pm 5\frac{V}{\mu s}$	$5pF$	$2.1V$	$-1.3V$	$2.2V$	$-2.2V$	$5MHz$	$> 80dB$	$> 60°$	$24\frac{nV}{\sqrt{Hz}}$
Spec											

Table 2: Device Sizing and Quiescent Point Values

Mosfet	S	W [μm]	L [μm]	I [μA]	$\lvert V_{OV}\rvert$ [V]	$V_{DS/SD}$ [V]
M1	4.7	4.7	1	13.45	0.175	2.20
M2	4.7	4.7	1	13.45	0.175	2.20
M3	7.2	7.2	1	13.45	0.298	1.20
M4	7.2	7.2	1	13.45	0.298	1.20
M5	4.05	4.05	1	26.90	0.268	1.61
M6	48.60	260	5.35	90.46	0.296	2.50
M7	13.50	13.50	1	90.46	0.268	2.50
M8	0.15	0.15	1	0.99	0.268	0.98
M9	10.5	10.5	1	0.99	0.290	0
M10	0.55	0.55	1	0.99	0.290	2.83
M11	0.55	0.55	1	0.99	0.293	1.20
M12	0.55	0.55	1	0.99	0.293	1.20
M13	0.55	0.55	1	0.99	0.293	1.19
M14	0.60	0.60	1	0.99	0.133	0.247
Other components:	$C_C = 4.4$ [pF]		$R_B = 135.5$ [kΩ]			

**Note: Values reflect the final optimized circuit.*

2. Schematic

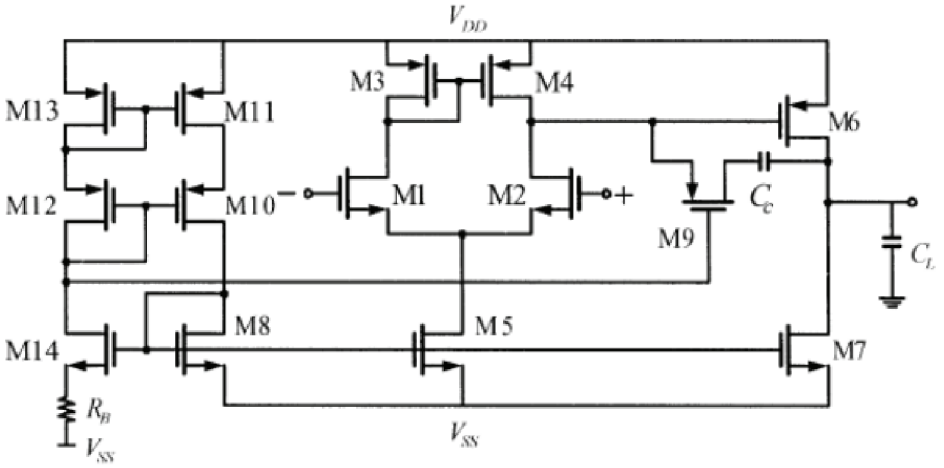


Figure 1: Schematic of the Two-Stage Operational Amplifier

3. Project Objectives

During the design of the operational amplifier, in addition to meeting the electrical specifications, three key criteria were applied to ensure an efficient and manufacturable circuit:

- **Minimization of Silicon Area:** The transistor dimensions (W and L) were kept as small as allowed by technology and performance limits, reducing layout size and parasitic capacitances, thus improving dynamic behavior.
- **Minimization of Power Consumption:** Static current was minimized through careful tuning of the bias network and current mirrors, ensuring that the differential pair and output stage operate with the lowest current required to meet SR and GBW specifications.
- **Manufacturing Layout Compatibility:** To ensure full compliance with foundry design rules, all calculated geometric dimensions were rounded to the nearest multiple of 50 nm.

4. Design Procedure and Calculations

4.1. V_{ov3} and C_C Determination

$$V_{in,CM}^{max} = V_{DD} - V_{ov3} - |V_{tp}| + V_{tn}$$
$$V_{ov3} = V_{DD} - V_{in,CM}^{max} - |V_{tp}| + V_{tn} = 0.209 \text{ V}$$
$$C_C = \frac{16}{3} kT \frac{1}{\omega_{0,dB}} \left(1 + \frac{SR}{\omega_{0,dB} V_{ov3}}\right) \frac{1}{\frac{v_{in,N}^2}{\Delta f}} = 2.15 \text{ pF}$$

4.2. Quiescent Currents and M6 Sizing

$$I_5 = C_C \cdot SR = 10.75 \ \mu\text{A}$$
$$I_7 = \frac{SR}{C_L + C_C} = 35.75 \ \mu\text{A} \quad (I_6 = I_7)$$
$$L_6 = C_c \sqrt{\frac{3}{2} \mu_p (V_{DD} - V_o^{max}) \frac{1}{\tan(PM)(C_c + C_L)\beta_n' S_1 V_{ov1}}} = 5.35 \ \mu\text{m}$$
$$V_{ov6} = V_{DD} - V_o^{max} = 0.300 \text{ V}$$
$$S_6 = \frac{2I_6}{\beta_p' V_{ov6}^2} = 19.14 \implies W_6 = S_6 L_6 = 102.4 \ \mu\text{m}$$

4.3. Differential Pair (M1, M2)

$$V_{ov1} = \frac{SR}{\omega_{0,dB}} = 0.159 \text{ V}$$
$$S_1 = S_2 = \frac{I_5}{\beta_n' V_{ov1}^2} = 2.33 \implies W_1 = W_2 = 2.33 \ \mu\text{m} \quad (L_{1,2} = 1 \ \mu\text{m})$$

4.4. Tail Current Sink (M5)

$$V_{ov5} = V_{in,CM}^{min} - V_{ov1} - V_{tn} - V_{SS} = 0.331 \text{ V}$$
$$S_5 = \frac{2I_5}{\beta_n' V_{ov5}^2} = 1.08 \implies W_5 = 1.08 \ \mu\text{m} \quad (L_5 = 1 \ \mu\text{m})$$

4.5. Output Current Sink (M7)

$$S_7 = S_5 \frac{I_7}{I_5} = 3.59 \implies W_7 = 3.59 \ \mu\text{m} \quad (L_7 = 1 \ \mu\text{m})$$

4.6. Active Load (M3, M4)

$$S_3 = S_4 = \frac{S_6}{2S_7} S_5 = 2.88 \implies W_3 = W_4 = 2.88\ \mu\text{m} \quad (L_{3,4} = 1\ \mu\text{m})$$

4.7. Compensation Resistor (M9)

$$g_{m6} = \frac{2I_6}{V_{ov6}} = 238.33\ \mu\text{A/V}$$

$$R_C = \frac{1}{g_{m6}} \left(1 + \frac{C_L}{C_C}\right) = 13.953\ \text{k}\Omega$$

$$S_9 = \frac{C_c}{C_C + C_L} S_6 = 5.75 \implies W_9 = 5.75\ \mu\text{m} \quad (L_9 = 1\ \mu\text{m})$$

$$V_{ov9} = \frac{1}{R_c \beta'_p S_9} = 0.300\ \text{V}$$

4.8. Biasing Network

$$I_{bias} = 1\ \mu\text{A}$$

$$S_{10...13} = S_6 \frac{I_{bias}}{I_6} = 0.55 \implies W_{10...13} = 0.55\ \mu\text{m} \quad (L_{10...13} = 1\ \mu\text{m})$$

$$S_8 = S_5 \frac{I_{bias}}{I_5} = 0.10 \implies W_8 = 0.10\ \mu\text{m} \quad (L_8 = 1\ \mu\text{m})$$

$$K = 4 \implies S_{14} = S_8 K = 0.40 \implies W_{14} = 0.40\ \mu\text{m} \quad (L_{14} = 1\ \mu\text{m})$$

$$V_{ov14} = \frac{V_{ov8}}{\sqrt{K}}$$

$$R_B = \frac{V_{ov8} - V_{ov14}}{I_{bias}} = 165.5\ \text{k}\Omega$$

4.9. Optimization

During the simulation phase, iterative refinements were applied to meet the electrical requirements while respecting the 50 nm manufacturing quantization grid. To satisfy the negative Slew Rate (SR–) specification, the size of M7 was increased, which in turn required a corresponding recalibration of the output stage and the active load to preserve systematic offset balance:

$$S_{7,new} = 6.7 \implies W_7 = 6.7\ \mu\text{m}$$

$$S_{6,new} = S_{6,old} \frac{I_{6,new}}{I_{6,old}} = 24.03 \implies W_6 = 128\ \mu\text{m}$$

$$S_3 = S_4 = S_5 \frac{S_{6,new}}{2S_{7,new}} = 2.95 \implies W_3 = W_4 = 2.95\ \mu\text{m}$$

To restore the Phase Margin (previously reduced due to the expanded bandwidth) the compensation zero was repositioned, and the transconductance of the differential pair was slightly decreased:

$$S_9 = 8.00$$

$$S_1 = S_2 = 2.30$$

To eliminate the saturation clipping observed during the negative output swing, the overdrive voltage of the output sink transistor (V_{ov7}) was lowered from 0.33V to 0.30V. To preserve the exact quiescent currents across all branches, a global area scaling factor ($K = 1.5$) was applied to the entire lower NMOS array, accompanied by a proportional adjustment of the bias resistor:

$$K > \left(\frac{V_{ov7,old}}{V_{ov7,new}}\right)^2 = \left(\frac{0.33}{0.30}\right)^2 = 1.29 \implies K = 1.5$$

$$S_{new} = S_{old} \cdot K \quad \text{for } \{\text{M5, M7, M8, M14}\}$$

$$S_{6,new} = S_{6,old}\frac{I_{6,new}}{I_{6,old}} = 35.87 \implies W_6 = 192.6\ \mu\text{m}$$

$$R_{B,new} = \frac{R_B}{\sqrt{K}} = 135.5\ \text{k}\Omega$$

To satisfy the strict high-frequency noise specification at 200 kHz, the compensation capacitor and differential stage gain was increased ($C_C = 4.4pF$,$S_{1,2} = 4.7$, $S_5 = 4.05$, $S_{3,4} = 7.2$). This upgrade delayed the crossover frequency to 10.5 MHz, severely degrading the phase margin to 40°. To restore system stability without further altering C_C, the compensation resistor was re-tuned by increasing W_9 to 10.5 μm, which successfully repositioned the compensation zero and recovered the required phase margin.

5. Simulation Outputs

5.1 Operating Point

The following log details the DC operating point parameters for all semiconductor devices in the circuit, confirming that the transistors are correctly biased, each one in the Saturation region except M9 that has to be in the Linear/Triode region.

```
Semiconductor Device Operating Points:
                        --- MOSFET Transistors ---
Name:            M2          M7          M8          M1          M5
Model:       nmod1cap    nmod1cap    nmod1cap    nmod1cap    nmod1cap
Id:          1.35e-05    6.70e-05    9.90e-07    1.35e-05    2.69e-05
Vgs:         8.93e-01    9.78e-01    9.78e-01    8.93e-01    9.78e-01
Vds:         2.20e+00    2.50e+00    9.78e-01    2.19e+00    1.61e+00
Vbs:        -1.61e+00    0.00e+00    0.00e+00   -1.61e+00    0.00e+00
Vth:         7.17e-01    7.10e-01    7.10e-01    7.17e-01    7.10e-01
Vdsat:       1.75e-01    2.68e-01    2.68e-01    1.75e-01    2.68e-01
Gm:          1.53e-04    5.00e-04    7.39e-06    1.53e-04    2.01e-04
Gds:         1.32e-07    6.54e-07    9.81e-09    1.32e-07    2.65e-07
Gmb:         5.16e-07    3.23e-06    4.77e-08    5.16e-07    1.30e-06
Cbd:         0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cbs:         0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgsov:       2.16e-16    4.60e-16    6.90e-18    2.16e-16    1.86e-16
Cgdov:       2.16e-16    4.60e-16    6.90e-18    2.16e-16    1.86e-16
Cgbov:       0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgs:         1.13e-14    2.40e-14    3.60e-16    1.13e-14    9.71e-15
Cgd:         0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgb:         0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00

Name:           M14         M6          M4          M11         M10
Model:       nmod1cap    pmod1cap    pmod1cap    pmod1cap    pmod1cap
Id:          9.90e-07    6.70e-05    1.35e-05    9.90e-07    9.90e-07
Vgs:         8.44e-01    1.30e+00   -2.50e-03    2.31e-03    1.63e+00
Vds:         2.47e+00    2.50e+00    1.20e+00    1.20e+00    2.83e+00
Vbs:        -1.34e-01    2.50e+00    1.20e+00    1.20e+00    4.02e+00
Vth:         7.11e-01   -9.01e-01   -9.01e-01   -9.01e-01   -9.07e-01
Vdsat:       1.33e-01   -2.96e-01   -2.98e-01   -2.93e-01   -2.90e-01
Gm:          1.49e-05    4.53e-04    9.02e-05    6.76e-06    6.82e-06
Gds:         9.66e-09    6.54e-07    1.33e-07    9.79e-09    9.63e-09
Gmb:         8.69e-08    2.92e-06    5.82e-07    4.37e-08    2.54e-08
Cbd:         0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cbs:         0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgsov:       2.76e-17    8.86e-15    3.31e-16    2.53e-17    2.53e-17
Cgdov:       2.76e-17    8.86e-15    3.31e-16    2.53e-17    2.53e-17
Cgbov:       0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgs:         1.44e-15    0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgd:         0.00e+00    2.47e-12    1.73e-14    1.32e-15    1.32e-15
Cgb:         0.00e+00    0.00e+00    0.00e+00    0.00e+00    0.00e+00

Name:           M13         M12         M3          M9
Model:       pmod1cap    pmod1cap    pmod1cap    pmod1cap
Id:          9.90e-07    9.90e-07    1.35e-05   -4.22e-08
Vgs:         0.00e+00    0.00e+00    0.00e+00   -1.20e+00
Vds:         1.19e+00    1.20e+00    1.20e+00   -3.34e-04
Vbs:         1.19e+00    2.39e+00    1.20e+00    1.20e+00
Vth:        -9.01e-01   -9.07e-01   -9.01e-01   -9.07e-01
Vdsat:      -2.93e-01   -2.93e-01   -2.98e-01   -2.90e-01
Gm:          6.76e-06    6.76e-06    9.02e-05    1.46e-07
Gds:         9.79e-09    9.79e-09    1.33e-07    1.26e-04
Gmb:         4.37e-08    2.53e-08    5.82e-07    5.43e-10
Cbd:         0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cbs:         0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgsov:       2.53e-17    2.53e-17    3.31e-16    4.83e-16
Cgdov:       2.53e-17    2.53e-17    3.31e-16    4.83e-16
Cgbov:       0.00e+00    0.00e+00    0.00e+00    0.00e+00
Cgs:         0.00e+00    0.00e+00    0.00e+00    1.89e-14
Cgd:         1.32e-15    1.32e-15    1.73e-14    1.89e-14
Cgb:         0.00e+00    0.00e+00    0.00e+00    0.00e+00
```

5.2 AC Sweep

To measure the open-loop AC response without introducing DC saturation from inherent offsets, a dedicated testbench was used. A 1GH inductor was inserted in the feedback path between the output and the inverting input (IN–), while a 1GF capacitor was connected from IN– to ground. At DC, the inductor behaves as a short and the capacitor as an open circuit, ensuring closed-loop stability and establishing the correct quiescent point. At high frequencies, the inductor opens and the capacitor effectively grounds IN–, breaking the loop and enabling an accurate open-loop AC sweep. This configuration is only used for the AC Sweep Analysis, it's not the final schematic, since it won't work with the other simulations.

The simulation results confirm that the optimized amplifier satisfies all stability requirements. The circuit achieves a DC gain of 106dB, with a dominant pole near 23Hz. The 0dB crossover occurs at 7MHz, where the phase reaches –120°, corresponding to a Phase Margin of more than 65°.

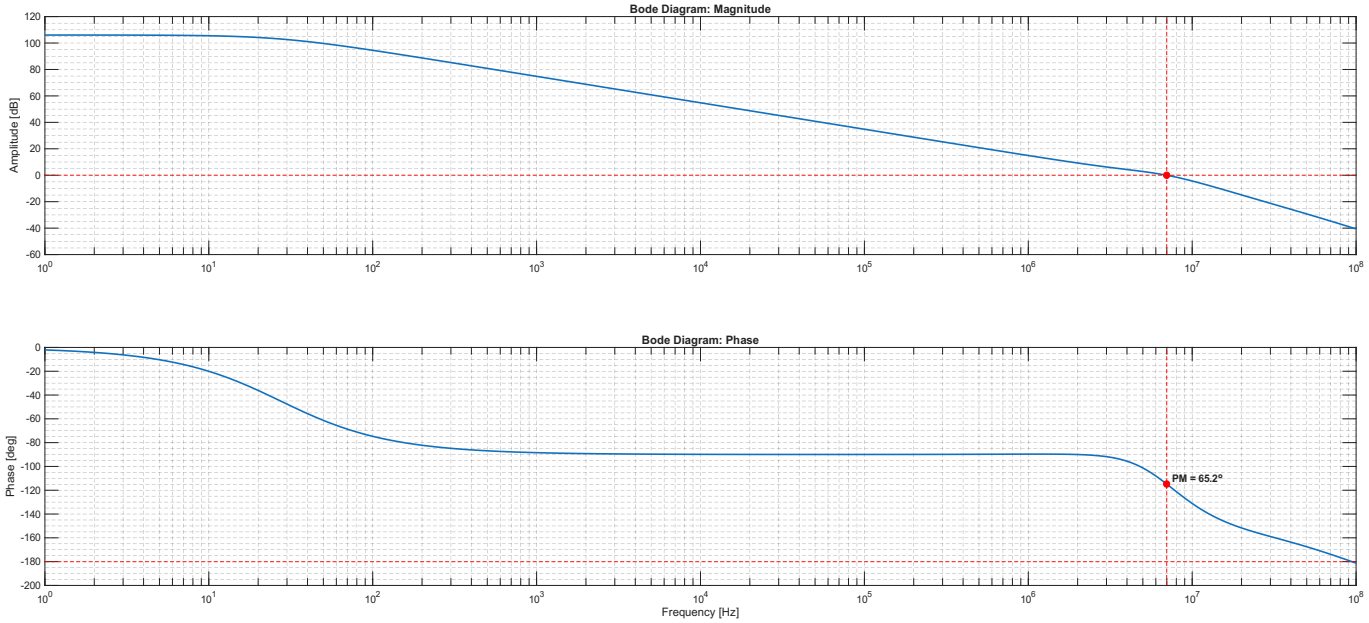


Figure 2: Amplitude-Phase Bode Diagram

5.3 DC Sweep IN

To verify the input common-mode voltage ($V_{in,CM}$) swing, a DC sweep analysis was performed in a closed-loop configuration. As shown in the simulation plot below, the valid linear operating region extends from -1.78V up to over 2.1V, fully satisfying the design specifications.

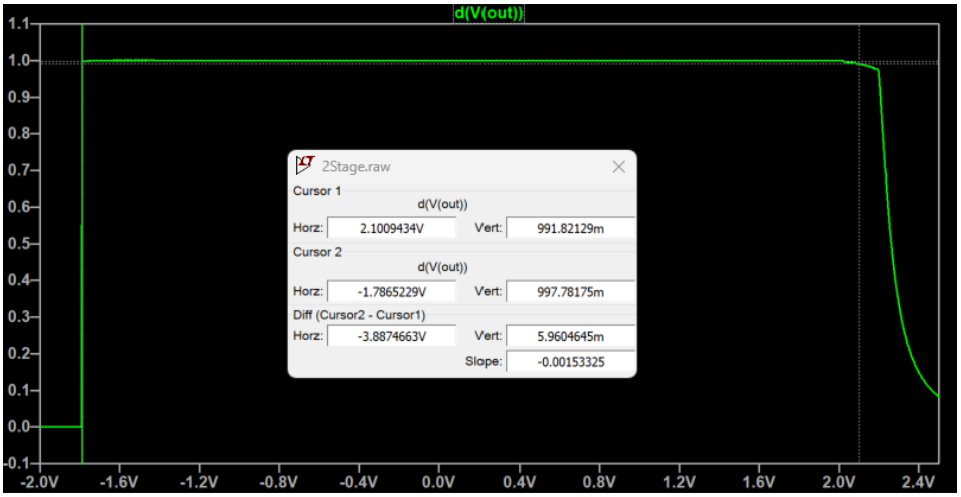


Figure 3: Derivative of the output voltage, highlighting the linear $V_{in,CM}$ boundaries.

5.4 DC Sweep OUT

To evaluate the output voltage boundaries (V_o^{max} and V_o^{min}), a DC sweep analysis was performed using an inverting amplifier configuration. Resistors of 1 MΩ and 10 MΩ were utilized to set a constant closed-loop gain of -10 while ensuring negligible current draw, thus preventing the output stage from being loaded. As demonstrated in the plot below, the amplifier successfully maintains its full constant gain across the required ±2.2 V output range (corresponding to an input sweep of ∓220 mV), fully satisfying the design specifications.

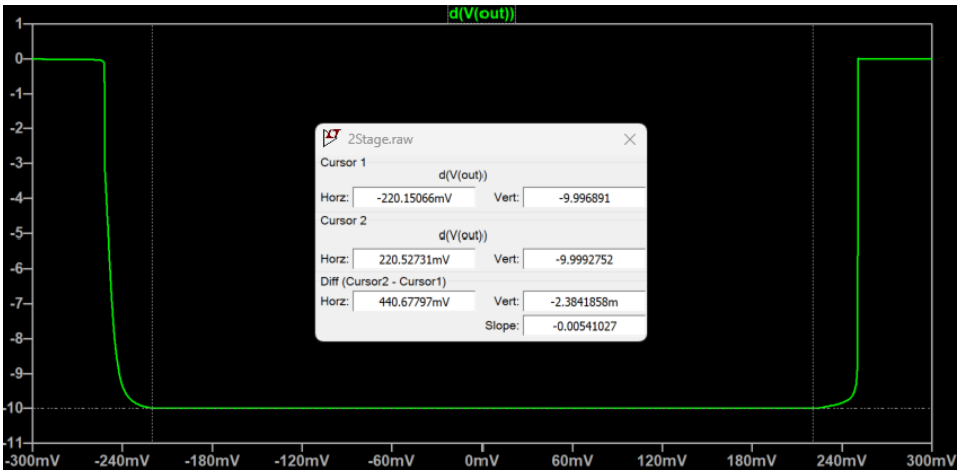


Figure 4: Derivative of the output voltage in the inverting configuration, confirming linear operation and constant gain over the required output swing.

5.5 Transient Analysis

To evaluate the large-signal dynamic response, a transient analysis was performed by applying a steep input step (`PULSE(0 1 10n 1n 1n 1u 2u)`). The Slew Rate (SR) was extracted by measuring the slope of the output waveform strictly between the 10% and 90% transition points. The final optimized values successfully meet the required ±5 V/μs specification.

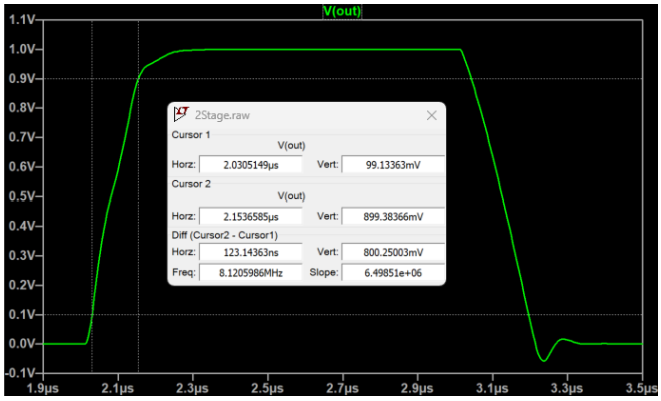


Figure 5: Positive Slew Rate Analysis

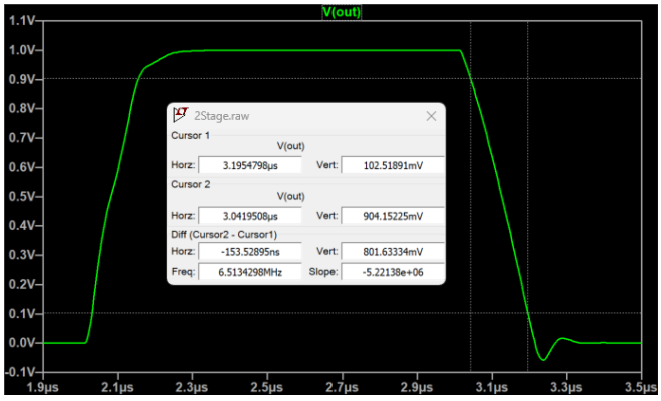


Figure 6: Negative Slew Rate Analysis

5.6 Noise analysis

An input-referred noise simulation (`.noise`) was performed to verify the 24 nV/$\sqrt{\text{Hz}}$ specification. In the mid-band frequency range, the thermal noise floor remains strictly below the required threshold, validating the correct sizing of the input differential pair and the compensation capacitor. The absence of a low-frequency Flicker ($1/f$) noise corner is expected, as the provided SPICE models do not include $1/f$ coefficients.

As frequency increases, the plotted input-referred noise rises, crossing the 24 nV/$\sqrt{\text{Hz}}$ boundary at approximately 270 kHz. This behavior is a natural consequence of the open-loop evaluation setup. The simulator calculates the input-referred noise by dividing the output noise by the frequency-dependent open-loop gain $|A(jf)|$. Since the amplifier's gain rolls off at -20 dB/dec after the dominant pole, the mathematical division by a decreasing gain causes the calculated input-referred noise to steeply increase at higher frequencies:

$$v_{in,N}(f) = \frac{v_{out,N}(f)}{|A(jf)|}$$

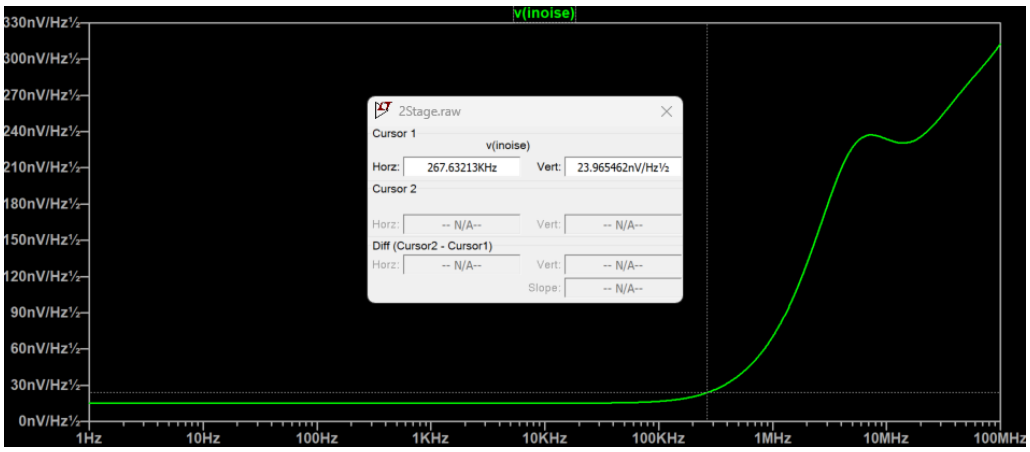


Figure 7: Noise Analysis, highlighting the maximum noise threshold.

5.7 Power Consumption

To minimize power consumption, the quiescent currents were constrained to a low-current operating point value that still satisfied the Slew Rate (SR) and Gain-Bandwidth Product (GBW) requirements. Both the differential input stage and the output stage were dimensioned to drive their respective capacitive loads efficiently, while the bias network was refined to operate at only $1\mu A$ per branch.

Table3 reports the theoretical static current allocation across the four principal active branches, together with the final total current and power obtained from the operating-point simulation:

Circuit Branch	**Current [μA]**
Differential Stage (I_{DIFF})	26.90
Output Stage (I_{CS})	67.01
Biasing Network 1 (I_{bias1})	0.99
Biasing Network 2 (I_{bias2})	0.99
Total Current (I_{tot})	**95.89**
Total Power (P_{tot}) [μW]	**479.5**

6. Final Performance Summary

Table 3 compares the initial minimum design specifications with the final simulated performance of the optimized operational amplifier.

Table 3: Comparison of Target Specifications and Achieved Performance

Parameter	**Unit**	**Target**	**Achieved Value**
SR^+	$[\frac{V}{\mu s}]$	$+5$	$+6.5$
SR^-	$[\frac{V}{\mu s}]$	-5	-5.2
$V_{cm,min}$	$[V]$	-1.3	-1.78
$V_{cm,max}$	$[V]$	2.1	2.20
$V_{o,min}$	$[V]$	-2.2	-2.23
$V_{o,max}$	$[V]$	2.2	2.20
f_{0db}	$[MHz]$	5	7.00
DC_{Gain}	$[dB]$	> 80	106
PM	$[^\circ]$	> 60	65.2
$\frac{v_{in,N}(f)}{\sqrt{\Delta f}}$	$[\frac{nV}{\sqrt{Hz}}]$	24	20
Power	$[\mu W]$	Minimize	479.5

7. Conclusion

The two-stage CMOS operational amplifier was fully designed, biased, and progressively refined to meet all the required electrical constraints. Through systematic adjustments of the dimension ratios in the active loads and biasing networks, the initial discrepancies observed under transient pulse excitation and slew-rate stress were successfully resolved. The final device sizing, rigorously aligned with the 50nm manufacturing quantization grid, delivers a stable and well-balanced compromise between dynamic performance, minimized silicon area, and low static power consumption, thereby achieving the central objectives of the project.